ECMO-PDU

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Portable ECMO Power Distribution Unit

Medical Critical-Care Transport / Redundant Power | Rev A1 - Documentation Baseline Program Date TBC | Architecture, Atomic Requirements, Interfaces & Verification

Mission profile. Intelligent redundant power manager for portable ECMO transport systems. Dual lithium-ion batteries and AC mains are hot-swapped so the blood-pump rail retains highest priority, with secondary displays and convenience loads shed first when energy or source capability is constrained.

Engineering baseline. The architecture includes medical isolation, 24 V motor-load distribution, source arbitration, battery health/cycle/impedance tracking, an isolated logging interface and a high-impact IP54 transport chassis. The supplied baseline states that no single point of failure may halt the blood pump and requires rigorous switchover/drop validation.

Battery packs, mains adapter, OR-ing devices, charger, pump interface, enclosure, isolation barrier, firmware and transport installation are controlled variants. The baseline phrase IEC 60601-1 Class III compliant requires program clarification of regulatory class versus electrical protection classification before final release; exact switchover, battery, leakage and environmental limits remain controlled parameters.

Definition	Controlled product interpretation
Product class	Portable ECMO Power Distribution Unit.
Safety / assurance	Intelligent redundant power manager for portable ECMO transport systems.
Documentation baseline	Program Date TBC - controlled PRD baseline; if marked TBC, program approval is required before external release.
Primary architecture	The architecture includes medical isolation, 24 V motor-load distribution, source arbitration, battery health/cycle/impedance tracking, an isolated logging interface and a high-impact IP54 transport chassis. The supplied baseline states that no single point of failure may halt the blood pump and requires rigorous switchover/drop validation.
Release philosophy	Battery packs, mains adapter, OR-ing devices, charger, pump interface, enclosure, isolation barrier, firmware and transport installation are controlled variants. The baseline phrase IEC 60601-1 Class III compliant requires program clarification of regulatory class versus electrical protection classification before final release; exact switchover, battery, leakage and environmental limits remain controlled parameters.

PRODUCT GOVERNANCE, PUMP-POWER PRIORITY & REQUIREMENT QUALITY

Product Governance, Pump-Power Priority & Requirement Quality is controlled as a distinct product mechanism. The PDU shall place ECMO blood-pump power above secondary displays and nonessential loads in every released source and fault state.

The credible failure concern is a secondary function competes with pump power during constrained energy or source fault. The allocated response is to shed or inhibit the lower-priority load and preserve the validated pump-power path.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-001	The PDU shall place ECMO blood-pump power above secondary displays and nonessential loads in every released source and fault state.	power-priority architecture, state-machine review and constrained-energy DVP
ECMOPDU-002	If a secondary function competes with pump power during constrained energy or source fault, the system shall shed or inhibit the lower-priority load and preserve the validated pump-power path.	power-priority architecture, state-machine review and constrained-energy DVP
ECMOPDU-003	A change to load priority, pump interface, secondary display, software policy or distribution topology shall reopen power-priority architecture, state-machine review and constrained-energy DVP before production release.	power-priority architecture, state-machine review and constrained-energy DVP

Verification, margin and future-change rule

Future variants affecting load priority, pump interface, secondary display, software policy or distribution topology are screened against the mechanism on this page. The release owner may retain prior evidence only where power-priority architecture, state-machine review and constrained-energy DVP confirms the similarity boundary.

SYSTEM BOUNDARY, ECMO PUMP / BATTERY / MAINS OWNERSHIP

The released architecture assigns explicit ownership to system boundary, ecmo pump / battery / mains ownership. The PDU shall define ownership of AC mains, dual batteries, chargers, pump rail, secondary rails, alarms and isolated logging interfaces.

A release review shall consider source-good, pump-ready or load-shed ownership is ambiguous across connected equipment; normal service is not assumed when that condition is present, and the product must hold the ambiguous transition, preserve the safest pump-power state and expose the interface fault.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-004	The PDU shall define ownership of AC mains, dual batteries, chargers, pump rail, secondary rails, alarms and isolated logging interfaces.	system boundary diagram, ICD and power-authority table
ECMOPDU-005	Detection of source-good, pump-ready or load-shed ownership is ambiguous across connected equipment shall cause the system to hold the ambiguous transition, preserve the safest pump-power state and expose the interface fault.	system boundary diagram, ICD and power-authority table
ECMOPDU-006	Evidence reuse after ECMO pump interface, battery pack, mains adapter, charger, display or logging system is permitted only when system boundary diagram, ICD and power-authority table demonstrates that the affected assumptions remain valid.	system boundary diagram, ICD and power-authority table

Verification, margin and future-change rule

The change trigger for this page is ECMO pump interface, battery pack, mains adapter, charger, display or logging system. The impact record identifies moved assumptions and the exact system boundary diagram, ICD and power-authority table needed for closure.

OPERATING MODES, TRANSPORT / ICU / SERVICE STATES

For operating modes, transport / icu / service states, the design intent is translated into an auditable interface and state model. Controlled modes shall cover mains operation, battery operation, source transfer, low-energy priority, transport, fault containment, service and controlled shutdown.

Verification focuses on the point where a mode prerequisite or transition state is invalid. The required containment is to remain on the safest available pump-power source and block nonessential transitions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-007	Controlled modes shall cover mains operation, battery operation, source transfer, low-energy priority, transport, fault containment, service and controlled shutdown.	state-machine verification, transition fault injection and restart testing
ECMOPDU-008	The design shall detect a mode prerequisite or transition state is invalid and shall remain on the safest available pump-power source and block nonessential transitions.	state-machine verification, transition fault injection and restart testing
ECMOPDU-009	Transport workflow, source logic, service mode, low-energy policy or shutdown sequence shall be treated as a controlled engineering change; approval requires state-machine verification, transition fault injection and restart testing.	state-machine verification, transition fault injection and restart testing

Verification, margin and future-change rule

Changes involving transport workflow, source logic, service mode, low-energy policy or shutdown sequence are engineering events, not editorial updates; approval depends on the targeted state-machine verification, transition fault injection and restart testing listed here.

AC MAINS INPUT, PROTECTION & MEDICAL POWER BOUNDARY

This page closes the product-level behavior for ac mains input, protection & medical power boundary. The mains path shall provide the protective and isolation functions allocated by the medical electrical safety concept before it supplies charger or DC distribution stages.

The mechanism is incomplete unless it addresses mains input, protection or isolation status is unsafe or unknown and can disconnect or ignore the affected mains path and continue from qualified battery power when available without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-010	The mains path shall provide the protective and isolation functions allocated by the medical electrical safety concept before it supplies charger or DC distribution stages.	mains fault DVP, dielectric/leakage evidence and transfer test
ECMOPDU-011	Loss or corruption involving mains input, protection or isolation status is unsafe or unknown shall not permit silent normal operation; the system shall disconnect or ignore the affected mains path and continue from qualified battery power when available.	mains fault DVP, dielectric/leakage evidence and transfer test
ECMOPDU-012	Production equivalence following mains module, cable, fuse, inlet, isolation barrier or protective-earth concept requires closure of mains fault DVP, dielectric/leakage evidence and transfer test.	mains fault DVP, dielectric/leakage evidence and transfer test

Verification, margin and future-change rule

For mains module, cable, fuse, inlet, isolation barrier or protective-earth concept, the configuration owner documents evidence reuse, new risks and the required mains fault DVP, dielectric/leakage evidence and transfer test before sign-off.

DUAL LITHIUM-ION BATTERY ARCHITECTURE

Dual Lithium-Ion Battery Architecture is treated as an engineering boundary rather than a descriptive feature. The PDU shall independently identify, monitor and qualify both lithium-ion packs before accepting either as a pump-power source.

A battery reports invalid identity, unsafe voltage/temperature or contradictory health data is treated as a diagnosable abnormal condition. The system response is to exclude that pack from source arbitration while preserving the other qualified source.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-013	The PDU shall independently identify, monitor and qualify both lithium-ion packs before accepting either as a pump-power source.	battery interface test, invalid-pack cases and source arbitration DVP
ECMOPDU-014	When a battery reports invalid identity, unsafe voltage/temperature or contradictory health data is present, the product shall exclude that pack from source arbitration while preserving the other qualified source.	battery interface test, invalid-pack cases and source arbitration DVP
ECMOPDU-015	The released configuration shall not be extended across battery chemistry, pack electronics, connector, communication or qualification criteria without battery interface test, invalid-pack cases and source arbitration DVP.	battery interface test, invalid-pack cases and source arbitration DVP

Verification, margin and future-change rule

Any modification to battery chemistry, pack electronics, connector, communication or qualification criteria is accepted only after the affected assumptions are rechecked and battery interface test, invalid-pack cases and source arbitration DVP is closed.

HOT-SWAP OR-ING, REVERSE BLOCKING & SOURCE ISOLATION

The product baseline allocates measurable behavior to hot-swap or-ing, reverse blocking & source isolation. The dual-battery/mains architecture shall prevent uncontrolled cross-current and shall support source removal/insertion without interruption beyond the pump tolerance.

The design prevents silent continuation when OR-ing device, source-isolation path or hot-swap control fails; it must instead isolate the faulty source path and maintain the remaining qualified pump-power path if physically available.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-016	The dual-battery/mains architecture shall prevent uncontrolled cross-current and shall support source removal/insertion without interruption beyond the pump tolerance.	hot-swap test, reverse-current test and fault injection
ECMOPDU-017	The released implementation shall isolate the faulty source path and maintain the remaining qualified pump-power path if physically available whenever OR-ing device, source-isolation path or hot-swap control fails.	hot-swap test, reverse-current test and fault injection
ECMOPDU-018	When OR-ing device, ideal-diode controller, contactor, FET, source impedance or harness occurs, the configuration record shall identify affected assumptions and close hot-swap test, reverse-current test and fault injection.	hot-swap test, reverse-current test and fault injection

Verification, margin and future-change rule

The release baseline remains valid across OR-ing device, ideal-diode controller, contactor, FET, source impedance or harness only within demonstrated similarity. Otherwise, hot-swap test, reverse-current test and fault injection is repeated or updated.

BATTERY CHARGER PATHS & CHARGE AUTHORITY

Battery Charger Paths & Charge Authority is controlled as a distinct product mechanism. Battery charging shall be electrically separated from pump-priority decisions and shall not reduce pump availability below the released margin.

The credible failure concern is charger fault, overcurrent or thermal limit threatens the DC distribution bus. The allocated response is to disable or derate charging before pump power is compromised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-019	Battery charging shall be electrically separated from pump-priority decisions and shall not reduce pump availability below the released margin.	charger fault test, load-priority DVP and thermal analysis
ECMOPDU-020	If charger fault, overcurrent or thermal limit threatens the DC distribution bus, the system shall disable or derate charging before pump power is compromised.	charger fault test, load-priority DVP and thermal analysis
ECMOPDU-021	A change to charger, charge-current policy, battery pack, thermal design or mains capacity shall reopen charger fault test, load-priority DVP and thermal analysis before production release.	charger fault test, load-priority DVP and thermal analysis

Verification, margin and future-change rule

Future variants affecting charger, charge-current policy, battery pack, thermal design or mains capacity are screened against the mechanism on this page. The release owner may retain prior evidence only where charger fault test, load-priority DVP and thermal analysis confirms the similarity boundary.

DC DISTRIBUTION & PRIMARY PUMP RAIL

The released architecture assigns explicit ownership to dc distribution & primary pump rail. The primary pump rail shall have a controlled voltage/current envelope, protection path and hold-up/source-transfer behavior derived from the ECMO pump interface.

A release review shall consider pump-rail voltage, current or continuity falls outside the validated envelope; normal service is not assumed when that condition is present, and the product must invoke the fastest available source/protection response and issue a persistent high-priority alarm.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-022	The primary pump rail shall have a controlled voltage/current envelope, protection path and hold-up/source-transfer behavior derived from the ECMO pump interface.	pump-load DVP, transient capture and protection fault injection
ECMOPDU-023	Detection of pump-rail voltage, current or continuity falls outside the validated envelope shall cause the system to invoke the fastest available source/protection response and issue a persistent high-priority alarm.	pump-load DVP, transient capture and protection fault injection
ECMOPDU-024	Evidence reuse after pump connector, cable, rail converter, fuse, current limit or load profile is permitted only when pump-load DVP, transient capture and protection fault injection demonstrates that the affected assumptions remain valid.	pump-load DVP, transient capture and protection fault injection

Verification, margin and future-change rule

The change trigger for this page is pump connector, cable, rail converter, fuse, current limit or load profile. The impact record identifies moved assumptions and the exact pump-load DVP, transient capture and protection fault injection needed for closure.

SECONDARY RAILS, DISPLAY LOADS & LOAD SHEDDING

For secondary rails, display loads & load shedding, the design intent is translated into an auditable interface and state model. Secondary loads shall be partitioned so their faults or peak demand cannot remove power from the blood pump.

Verification focuses on the point where secondary-rail overload, short or low-energy policy is active. The required containment is to disconnect or shed the affected secondary load while keeping pump distribution energized.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-025	Secondary loads shall be partitioned so their faults or peak demand cannot remove power from the blood pump.	short-circuit/load-shed testing and rail-isolation review
ECMOPDU-026	The design shall detect secondary-rail overload, short or low-energy policy is active and shall disconnect or shed the affected secondary load while keeping pump distribution energized.	short-circuit/load-shed testing and rail-isolation review
ECMOPDU-027	Display, accessory rail, USB/service load, converter or load-shed policy shall be treated as a controlled engineering change; approval requires short-circuit/load-shed testing and rail-isolation review.	short-circuit/load-shed testing and rail-isolation review

Verification, margin and future-change rule

Changes involving display, accessory rail, USB/service load, converter or load-shed policy are engineering events, not editorial updates; approval depends on the targeted short-circuit/load-shed testing and rail-isolation review listed here.

24 V MOTOR-LOAD INTERFACE & TRANSIENT BEHAVIOR

This page closes the product-level behavior for 24 v motor-load interface & transient behavior. The 24 V DC motor-load interface shall tolerate the released pump inrush, commutation and regenerative/transient behavior without destabilizing source arbitration.

The mechanism is incomplete unless it addresses motor-load transient drives the bus outside the controlled pump envelope and can use the validated hold-up/current-limit/source response while preventing reset of the pump-priority controller without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-028	The 24 V DC motor-load interface shall tolerate the released pump inrush, commutation and regenerative/transient behavior without destabilizing source arbitration.	motor-load transient DVP, impedance analysis and bus capture
ECMOPDU-029	Loss or corruption involving motor-load transient drives the bus outside the controlled pump envelope shall not permit silent normal operation; the system shall use the validated hold-up/current-limit/source response while preventing reset of the pump-priority controller.	motor-load transient DVP, impedance analysis and bus capture
ECMOPDU-030	Production equivalence following pump motor, 24 V rail, cable, converter, filter or hold-up network requires closure of motor-load transient DVP, impedance analysis and bus capture.	motor-load transient DVP, impedance analysis and bus capture

Verification, margin and future-change rule

For pump motor, 24 V rail, cable, converter, filter or hold-up network, the configuration owner documents evidence reuse, new risks and the required motor-load transient DVP, impedance analysis and bus capture before sign-off.

SOURCE-TRANSFER STATE MACHINE & SEAMLESS SWITCHOVER

Source-Transfer State Machine & Seamless Switchover is treated as an engineering boundary rather than a descriptive feature. Source arbitration shall select and transfer among mains and dual batteries using qualified source status, priority and anti-chatter logic.

The active source degrades or is removed is treated as a diagnosable abnormal condition. The system response is to transfer to a qualified alternate source within the pump continuity requirement and record the transition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-031	Source arbitration shall select and transfer among mains and dual batteries using qualified source status, priority and anti-chatter logic.	mains/battery removal tests, transfer-timing measurement and anti-chatter test
ECMOPDU-032	When the active source degrades or is removed is present, the product shall transfer to a qualified alternate source within the pump continuity requirement and record the transition.	mains/battery removal tests, transfer-timing measurement and anti-chatter test
ECMOPDU-033	The released configuration shall not be extended across source priority, thresholds, debounce, transfer controller or source-good signal without mains/battery removal tests, transfer-timing measurement and anti-chatter test.	mains/battery removal tests, transfer-timing measurement and anti-chatter test

Verification, margin and future-change rule

Any modification to source priority, thresholds, debounce, transfer controller or source-good signal is accepted only after the affected assumptions are rechecked and mains/battery removal tests, transfer-timing measurement and anti-chatter test is closed.

LOW-ENERGY PRIORITIZATION & SURVIVAL POLICY

The product baseline allocates measurable behavior to low-energy prioritization & survival policy. When stored energy becomes constrained, the PDU shall preserve pump power, source supervision and critical alarms ahead of secondary functions.

The design prevents silent continuation when remaining energy crosses the controlled reserve threshold; it must instead shed nonessential loads and present persistent reserve/replace-source indication while maintaining pump support.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-034	When stored energy becomes constrained, the PDU shall preserve pump power, source supervision and critical alarms ahead of secondary functions.	battery depletion DVP, load-priority test and alarm validation
ECMOPDU-035	The released implementation shall shed nonessential loads and present persistent reserve/replace-source indication while maintaining pump support whenever remaining energy crosses the controlled reserve threshold.	battery depletion DVP, load-priority test and alarm validation
ECMOPDU-036	When reserve threshold, load priority, alarm policy, battery capacity or pump load occurs, the configuration record shall identify affected assumptions and close battery depletion DVP, load-priority test and alarm validation.	battery depletion DVP, load-priority test and alarm validation

Verification, margin and future-change rule

The release baseline remains valid across reserve threshold, load priority, alarm policy, battery capacity or pump load only within demonstrated similarity. Otherwise, battery depletion DVP, load-priority test and alarm validation is repeated or updated.

BATTERY SOC / SOH / IMPEDANCE ESTIMATION

Battery Soc / Soh / Impedance Estimation is controlled as a distinct product mechanism. Battery health monitoring shall combine pack-provided data and controlled local measurements needed for source qualification and service decisions.

The credible failure concern is SOC, SOH or impedance data is stale, contradictory or unavailable. The allocated response is to use the conservative source-qualification policy and avoid claiming unsupported runtime remaining time.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-037	Battery health monitoring shall combine pack-provided data and controlled local measurements needed for source qualification and service decisions.	battery correlation test, stale-data injection and service-threshold review
ECMOPDU-038	If SOC, SOH or impedance data is stale, contradictory or unavailable, the system shall use the conservative source-qualification policy and avoid claiming unsupported runtime remaining time.	battery correlation test, stale-data injection and service-threshold review
ECMOPDU-039	A change to battery BMS, impedance method, SOC algorithm, sensor or communication interface shall reopen battery correlation test, stale-data injection and service-threshold review before production release.	battery correlation test, stale-data injection and service-threshold review

Verification, margin and future-change rule

Future variants affecting battery BMS, impedance method, SOC algorithm, sensor or communication interface are screened against the mechanism on this page. The release owner may retain prior evidence only where battery correlation test, stale-data injection and service-threshold review confirms the similarity boundary.

BATTERY CYCLE COUNTING & USAGE HISTORY

The released architecture assigns explicit ownership to battery cycle counting & usage history. The lifecycle record shall accumulate validated battery cycles and relevant high-load/temperature exposure needed for replacement policy.

A release review shall consider cycle-history data is missing, reset unexpectedly or inconsistent with pack identity; normal service is not assumed when that condition is present, and the product must flag the pack history as uncertain and require service disposition before long-duration reliance.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-040	The lifecycle record shall accumulate validated battery cycles and relevant high-load/temperature exposure needed for replacement policy.	cycle-count audit, NVM fault injection and pack-replacement workflow test
ECMOPDU-041	Detection of cycle-history data is missing, reset unexpectedly or inconsistent with pack identity shall cause the system to flag the pack history as uncertain and require service disposition before long-duration reliance.	cycle-count audit, NVM fault injection and pack-replacement workflow test
ECMOPDU-042	Evidence reuse after cycle algorithm, NVM, pack replacement, battery identity or logging schema is permitted only when cycle-count audit, NVM fault injection and pack-replacement workflow test demonstrates that the affected assumptions remain valid.	cycle-count audit, NVM fault injection and pack-replacement workflow test

Verification, margin and future-change rule

The change trigger for this page is cycle algorithm, NVM, pack replacement, battery identity or logging schema. The impact record identifies moved assumptions and the exact cycle-count audit, NVM fault injection and pack-replacement workflow test needed for closure.

PACK REPLACEMENT CRITERIA & SERVICE LOCKOUT

For pack replacement criteria & service lockout, the design intent is translated into an auditable interface and state model. Mandatory pack replacement criteria shall be based on approved cycle, impedance, age and condition thresholds rather than calendar age alone.

Verification focuses on the point where a pack exceeds a replacement threshold or its service history is uncertain. The required containment is to exclude or restrict the pack according to the service policy and issue a replacement indication.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-043	Mandatory pack replacement criteria shall be based on approved cycle, impedance, age and condition thresholds rather than calendar age alone.	aged-pack test, threshold review and service workflow verification
ECMOPDU-044	The design shall detect a pack exceeds a replacement threshold or its service history is uncertain and shall exclude or restrict the pack according to the service policy and issue a replacement indication.	aged-pack test, threshold review and service workflow verification
ECMOPDU-045	Replacement threshold, battery supplier, impedance model, service tool or pack generation shall be treated as a controlled engineering change; approval requires aged-pack test, threshold review and service workflow verification.	aged-pack test, threshold review and service workflow verification

Verification, margin and future-change rule

Changes involving replacement threshold, battery supplier, impedance model, service tool or pack generation are engineering events, not editorial updates; approval depends on the targeted aged-pack test, threshold review and service workflow verification listed here.

BATTERY DISCONNECT, FUSING & FAULT CONTAINMENT

This page closes the product-level behavior for battery disconnect, fusing & fault containment. Each battery branch shall contain the protective/disconnect elements required to isolate an internal or downstream fault without disabling the healthy source.

The mechanism is incomplete unless it addresses battery branch short, fuse event or disconnect failure occurs and can isolate the affected branch and preserve the remaining qualified pump source when possible without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-046	Each battery branch shall contain the protective/disconnect elements required to isolate an internal or downstream fault without disabling the healthy source.	branch short-circuit test, fuse/selectivity analysis and disconnect fault injection
ECMOPDU-047	Loss or corruption involving battery branch short, fuse event or disconnect failure occurs shall not permit silent normal operation; the system shall isolate the affected branch and preserve the remaining qualified pump source when possible.	branch short-circuit test, fuse/selectivity analysis and disconnect fault injection
ECMOPDU-048	Production equivalence following fuse, contactor/FET, pack connector, harness or branch protection requires closure of branch short-circuit test, fuse/selectivity analysis and disconnect fault injection.	branch short-circuit test, fuse/selectivity analysis and disconnect fault injection

Verification, margin and future-change rule

For fuse, contactor/FET, pack connector, harness or branch protection, the configuration owner documents evidence reuse, new risks and the required branch short-circuit test, fuse/selectivity analysis and disconnect fault injection before sign-off.

MEDICAL ISOLATION BARRIERS & 24 V LOAD PARTITION

Medical Isolation Barriers & 24 V Load Partition is treated as an engineering boundary rather than a descriptive feature. Isolation barriers shall maintain the medical electrical separation allocated between mains, pump/load circuits, logging/service and accessible interfaces.

Barrier integrity or isolation monitoring cannot be established is treated as a diagnosable abnormal condition. The system response is to remove the affected path from service and block release/operation as allocated by the safety concept.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-049	Isolation barriers shall maintain the medical electrical separation allocated between mains, pump/load circuits, logging/service and accessible interfaces.	dielectric/leakage test, barrier review and isolation fault injection
ECMOPDU-050	When barrier integrity or isolation monitoring cannot be established is present, the product shall remove the affected path from service and block release/operation as allocated by the safety concept.	dielectric/leakage test, barrier review and isolation fault injection
ECMOPDU-051	The released configuration shall not be extended across isolator, DC/DC, PCB spacing, connector, logging interface or mains module without dielectric/leakage test, barrier review and isolation fault injection.	dielectric/leakage test, barrier review and isolation fault injection

Verification, margin and future-change rule

Any modification to isolator, DC/DC, PCB spacing, connector, logging interface or mains module is accepted only after the affected assumptions are rechecked and dielectric/leakage test, barrier review and isolation fault injection is closed.

LEAKAGE CURRENT, PROTECTIVE EARTH & APPLIED-PART BOUNDARY

The product baseline allocates measurable behavior to leakage current, protective earth & applied-part boundary. The program shall define the applicable IEC 60601-1 equipment/protection classification and demonstrate leakage/protective measures for the actual ECMO system connection.

The design prevents silent continuation when required classification mapping or leakage/protective evidence is incomplete; it must instead block release of the affected configuration rather than relying on software mitigation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-052	The program shall define the applicable IEC 60601-1 equipment/protection classification and demonstrate leakage/protective measures for the actual ECMO system connection.	medical electrical safety report, leakage/earth tests and classification record
ECMOPDU-053	The released implementation shall block release of the affected configuration rather than relying on software mitigation whenever required classification mapping or leakage/protective evidence is incomplete.	medical electrical safety report, leakage/earth tests and classification record
ECMOPDU-054	When equipment class, applied-part boundary, grounding, enclosure or pump interface occurs, the configuration record shall identify affected assumptions and close medical electrical safety report, leakage/earth tests and classification record.	medical electrical safety report, leakage/earth tests and classification record

Verification, margin and future-change rule

The release baseline remains valid across equipment class, applied-part boundary, grounding, enclosure or pump interface only within demonstrated similarity. Otherwise, medical electrical safety report, leakage/earth tests and classification record is repeated or updated.

SINGLE-FAULT ARCHITECTURE & INDEPENDENCE

Single-Fault Architecture & Independence is controlled as a distinct product mechanism. The power architecture shall be evaluated so no single electrical fault identified by the risk analysis can remove all qualified pump-power sources.

The credible failure concern is a single fault defeats both source paths or a shared dependency removes pump power. The allocated response is to fail the design/release safety gate until the common dependency is eliminated or justified.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-055	The power architecture shall be evaluated so no single electrical fault identified by the risk analysis can remove all qualified pump-power sources.	single-fault analysis, fault-tree review and physical fault-injection evidence
ECMOPDU-056	If a single fault defeats both source paths or a shared dependency removes pump power, the system shall fail the design/release safety gate until the common dependency is eliminated or justified.	single-fault analysis, fault-tree review and physical fault-injection evidence
ECMOPDU-057	A change to shared controller, connector, rail, fuse, ground, firmware, OR-ing path or sensor shall reopen single-fault analysis, fault-tree review and physical fault-injection evidence before production release.	single-fault analysis, fault-tree review and physical fault-injection evidence

Verification, margin and future-change rule

Future variants affecting shared controller, connector, rail, fuse, ground, firmware, OR-ing path or sensor are screened against the mechanism on this page. The release owner may retain prior evidence only where single-fault analysis, fault-tree review and physical fault-injection evidence confirms the similarity boundary.

NO-SINGLE-POINT LOSS OF BLOOD-PUMP POWER

The released architecture assigns explicit ownership to no-single-point loss of blood-pump power. The baseline no-single-point-loss objective shall be traced to concrete source, distribution, control and connector mechanisms with objective evidence.

A release review shall consider any test or analysis demonstrates a single point capable of halting the pump; normal service is not assumed when that condition is present, and the product must withhold production release until the architecture or allocation is corrected.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-058	The baseline no-single-point-loss objective shall be traced to concrete source, distribution, control and connector mechanisms with objective evidence.	risk-control traceability, fault injection and end-to-end pump continuity test
ECMOPDU-059	Detection of any test or analysis demonstrates a single point capable of halting the pump shall cause the system to withhold production release until the architecture or allocation is corrected.	risk-control traceability, fault injection and end-to-end pump continuity test
ECMOPDU-060	Evidence reuse after pump rail, controller, connector, source path, distribution board or protection device is permitted only when risk-control traceability, fault injection and end-to-end pump continuity test demonstrates that the affected assumptions remain valid.	risk-control traceability, fault injection and end-to-end pump continuity test

Verification, margin and future-change rule

The change trigger for this page is pump rail, controller, connector, source path, distribution board or protection device. The impact record identifies moved assumptions and the exact risk-control traceability, fault injection and end-to-end pump continuity test needed for closure.

PUMP-POWER CONTINUITY MONITORING & FIRST-FAULT RESPONSE

For pump-power continuity monitoring & first-fault response, the design intent is translated into an auditable interface and state model. The PDU shall monitor pump-rail voltage/current and source state sufficiently to distinguish source transition, pump load change and distribution fault.

Verification focuses on the point where pump continuity degrades or source state becomes inconsistent. The required containment is to raise the highest-priority power alarm, preserve the best available source and capture first-fault evidence.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-061	The PDU shall monitor pump-rail voltage/current and source state sufficiently to distinguish source transition, pump load change and distribution fault.	pump-rail monitoring test, source-fault injection and first-fault log review
ECMOPDU-062	The design shall detect pump continuity degrades or source state becomes inconsistent and shall raise the highest-priority power alarm, preserve the best available source and capture first-fault evidence.	pump-rail monitoring test, source-fault injection and first-fault log review
ECMOPDU-063	Voltage/current sensor, threshold, source state, alarm logic or sampling shall be treated as a controlled engineering change; approval requires pump-rail monitoring test, source-fault injection and first-fault log review.	pump-rail monitoring test, source-fault injection and first-fault log review

Verification, margin and future-change rule

Changes involving voltage/current sensor, threshold, source state, alarm logic or sampling are engineering events, not editorial updates; approval depends on the targeted pump-rail monitoring test, source-fault injection and first-fault log review listed here.

VOLTAGE / CURRENT SENSING & SENSOR PLAUSIBILITY

This page closes the product-level behavior for voltage / current sensing & sensor plausibility. Critical source and pump measurements shall have range, plausibility and diagnostic checks appropriate to their use in power arbitration and alarms.

The mechanism is incomplete unless it addresses sensor offset, saturation, open circuit or contradictory readings occur and can exclude invalid data from source decisions and use the conservative fallback state without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-064	Critical source and pump measurements shall have range, plausibility and diagnostic checks appropriate to their use in power arbitration and alarms.	metrology WCCA, sensor fault injection and calibration verification
ECMOPDU-065	Loss or corruption involving sensor offset, saturation, open circuit or contradictory readings occur shall not permit silent normal operation; the system shall exclude invalid data from source decisions and use the conservative fallback state.	metrology WCCA, sensor fault injection and calibration verification
ECMOPDU-066	Production equivalence following sensor, ADC, shunt/Hall element, divider, reference or filter requires closure of metrology WCCA, sensor fault injection and calibration verification.	metrology WCCA, sensor fault injection and calibration verification

Verification, margin and future-change rule

For sensor, ADC, shunt/Hall element, divider, reference or filter, the configuration owner documents evidence reuse, new risks and the required metrology WCCA, sensor fault injection and calibration verification before sign-off.

THERMAL MANAGEMENT, BATTERY / POWER-STAGE LIMITS

Thermal Management, Battery / Power-Stage Limits is treated as an engineering boundary rather than a descriptive feature. Thermal design shall keep batteries, OR-ing devices, chargers and converters within approved limits across ICU and transport mission profiles.

Temperature exceeds a controlled derating/shutdown threshold or a critical sensor is invalid is treated as a diagnosable abnormal condition. The system response is to reduce charging/secondary loads first and preserve pump-power capability as long as safely possible.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-067	Thermal design shall keep batteries, OR-ing devices, chargers and converters within approved limits across ICU and transport mission profiles.	thermal DVP, worst-case load test and sensor-fault injection
ECMOPDU-068	When temperature exceeds a controlled derating/shutdown threshold or a critical sensor is invalid is present, the product shall reduce charging/secondary loads first and preserve pump-power capability as long as safely possible.	thermal DVP, worst-case load test and sensor-fault injection
ECMOPDU-069	The released configuration shall not be extended across battery, charger, converter, enclosure, airflow/conduction path or ambient profile without thermal DVP, worst-case load test and sensor-fault injection.	thermal DVP, worst-case load test and sensor-fault injection

Verification, margin and future-change rule

Any modification to battery, charger, converter, enclosure, airflow/conduction path or ambient profile is accepted only after the affected assumptions are rechecked and thermal DVP, worst-case load test and sensor-fault injection is closed.

STATE-MACHINE WATCHDOG, RESET & FAIL-SAFE STARTUP

The product baseline allocates measurable behavior to state-machine watchdog, reset & fail-safe startup. Runtime supervision shall detect stalled source arbitration, corrupted state and unsafe reset/startup sequencing.

The design prevents silent continuation when watchdog, reset or state-consistency check fails; it must instead default to the safest source hardware condition, keep pump priority active and expose the controller fault.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-070	Runtime supervision shall detect stalled source arbitration, corrupted state and unsafe reset/startup sequencing.	watchdog/reset fault injection, startup test and state-corruption cases
ECMOPDU-071	The released implementation shall default to the safest source hardware condition, keep pump priority active and expose the controller fault whenever watchdog, reset or state-consistency check fails.	watchdog/reset fault injection, startup test and state-corruption cases
ECMOPDU-072	When MCU, watchdog, reset IC, firmware, NVM state or startup sequence occurs, the configuration record shall identify affected assumptions and close watchdog/reset fault injection, startup test and state-corruption cases.	watchdog/reset fault injection, startup test and state-corruption cases

Verification, margin and future-change rule

The release baseline remains valid across MCU, watchdog, reset IC, firmware, NVM state or startup sequence only within demonstrated similarity. Otherwise, watchdog/reset fault injection, startup test and state-corruption cases is repeated or updated.

SOFTWARE TIMING, WCET & SOURCE-TRANSFER RESPONSE

Software Timing, Wcet & Source-Transfer Response is controlled as a distinct product mechanism. Source-control software shall meet controlled WCET and response-time budgets that are subordinate to physical pump-continuity requirements.

The credible failure concern is software execution exceeds its timing budget or misses a source event. The allocated response is to use available hardware fallback/hold-up and enter the diagnosed degraded state without hiding the timing failure.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-073	Source-control software shall meet controlled WCET and response-time budgets that are subordinate to physical pump-continuity requirements.	WCET analysis, stress test and delayed-task fault injection
ECMOPDU-074	If software execution exceeds its timing budget or misses a source event, the system shall use available hardware fallback/hold-up and enter the diagnosed degraded state without hiding the timing failure.	WCET analysis, stress test and delayed-task fault injection
ECMOPDU-075	A change to task period, scheduler, MCU load, algorithm or communication timing shall reopen WCET analysis, stress test and delayed-task fault injection before production release.	WCET analysis, stress test and delayed-task fault injection

Verification, margin and future-change rule

Future variants affecting task period, scheduler, MCU load, algorithm or communication timing are screened against the mechanism on this page. The release owner may retain prior evidence only where WCET analysis, stress test and delayed-task fault injection confirms the similarity boundary.

FAULT LATCHING, RECOVERY & SAFE RESTART

The released architecture assigns explicit ownership to fault latching, recovery & safe restart. Faults affecting pump-power confidence shall have controlled latching and recovery rules so transient clearing does not cause unstable source switching.

A release review shall consider a latched fault clears without the required validation or source stability window; normal service is not assumed when that condition is present, and the product must retain the safe source selection and require the defined acknowledgement/revalidation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-076	Faults affecting pump-power confidence shall have controlled latching and recovery rules so transient clearing does not cause unstable source switching.	fault-reset test, source bounce test and power-cycle verification
ECMOPDU-077	Detection of a latched fault clears without the required validation or source stability window shall cause the system to retain the safe source selection and require the defined acknowledgement/revalidation.	fault-reset test, source bounce test and power-cycle verification
ECMOPDU-078	Evidence reuse after fault manager, debounce, reset policy, source-good threshold or service acknowledgement is permitted only when fault-reset test, source bounce test and power-cycle verification demonstrates that the affected assumptions remain valid.	fault-reset test, source bounce test and power-cycle verification

Verification, margin and future-change rule

The change trigger for this page is fault manager, debounce, reset policy, source-good threshold or service acknowledgement. The impact record identifies moved assumptions and the exact fault-reset test, source bounce test and power-cycle verification needed for closure.

ISOLATED LOGGING INTERFACE

For isolated logging interface, the design intent is translated into an auditable interface and state model. The logging interface shall remain electrically and logically isolated from pump-power authority and shall not introduce a wireless attack surface.

Verification focuses on the point where logging host, cable or parser misbehaves. The required containment is to isolate or disable logging while leaving pump distribution and source arbitration unchanged.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-079	The logging interface shall remain electrically and logically isolated from pump-power authority and shall not introduce a wireless attack surface.	isolation test, malformed-log command test and interface review
ECMOPDU-080	The design shall detect logging host, cable or parser misbehaves and shall isolate or disable logging while leaving pump distribution and source arbitration unchanged.	isolation test, malformed-log command test and interface review
ECMOPDU-081	Logging port, isolator, service tool, protocol or cable shall be treated as a controlled engineering change; approval requires isolation test, malformed-log command test and interface review.	isolation test, malformed-log command test and interface review

Verification, margin and future-change rule

Changes involving logging port, isolator, service tool, protocol or cable are engineering events, not editorial updates; approval depends on the targeted isolation test, malformed-log command test and interface review listed here.

NO-WIRELESS ATTACK SURFACE & NETWORK MINIMIZATION

This page closes the product-level behavior for no-wireless attack surface & network minimization. The released PDU shall contain no enabled wireless radio; external connectivity shall be limited to the approved isolated service/logging interfaces.

The mechanism is incomplete unless it addresses an unintended wireless interface or unauthorized network service is present and can block release or disable the interface through controlled configuration before clinical use without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-082	The released PDU shall contain no enabled wireless radio; external connectivity shall be limited to the approved isolated service/logging interfaces.	hardware BOM review, RF scan, service enumeration and configuration audit
ECMOPDU-083	Loss or corruption involving an unintended wireless interface or unauthorized network service is present shall not permit silent normal operation; the system shall block release or disable the interface through controlled configuration before clinical use.	hardware BOM review, RF scan, service enumeration and configuration audit
ECMOPDU-084	Production equivalence following module BOM, MCU feature set, debug adapter, service interface or firmware requires closure of hardware BOM review, RF scan, service enumeration and configuration audit.	hardware BOM review, RF scan, service enumeration and configuration audit

Verification, margin and future-change rule

For module BOM, MCU feature set, debug adapter, service interface or firmware, the configuration owner documents evidence reuse, new risks and the required hardware BOM review, RF scan, service enumeration and configuration audit before sign-off.

SERVICE ACCESS, AUTHORIZATION & MAINTENANCE BOUNDARY

Service Access, Authorization & Maintenance Boundary is treated as an engineering boundary rather than a descriptive feature. Service actions shall require authorized access and shall not permit bypass of pump-priority, source protection or critical alarm mechanisms.

Service authentication fails or an unsafe privileged action is requested is treated as a diagnosable abnormal condition. The system response is to deny the action and retain normal pump-priority protections.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-085	Service actions shall require authorized access and shall not permit bypass of pump-priority, source protection or critical alarm mechanisms.	service-role test, negative authorization cases and maintenance workflow review
ECMOPDU-086	When service authentication fails or an unsafe privileged action is requested is present, the product shall deny the action and retain normal pump-priority protections.	service-role test, negative authorization cases and maintenance workflow review
ECMOPDU-087	The released configuration shall not be extended across service tool, credential, debug port, calibration command or maintenance role without service-role test, negative authorization cases and maintenance workflow review.	service-role test, negative authorization cases and maintenance workflow review

Verification, margin and future-change rule

Any modification to service tool, credential, debug port, calibration command or maintenance role is accepted only after the affected assumptions are rechecked and service-role test, negative authorization cases and maintenance workflow review is closed.

EVENT LOGGING, FIRST-FAULT CAPTURE & TRACEABILITY

The product baseline allocates measurable behavior to event logging, first-fault capture & traceability. The PDU shall record source transfers, low-energy events, battery health, pump-rail faults and service actions with a controlled first-fault strategy.

The design prevents silent continuation when logging capacity is exhausted or event records are inconsistent; it must instead preserve the configured critical-event subset and flag service attention without impacting pump power.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-088	The PDU shall record source transfers, low-energy events, battery health, pump-rail faults and service actions with a controlled first-fault strategy.	log rollover test, corruption injection and event-timestamp review
ECMOPDU-089	The released implementation shall preserve the configured critical-event subset and flag service attention without impacting pump power whenever logging capacity is exhausted or event records are inconsistent.	log rollover test, corruption injection and event-timestamp review
ECMOPDU-090	When log schema, NVM, retention policy, timestamp source or service tool occurs, the configuration record shall identify affected assumptions and close log rollover test, corruption injection and event-timestamp review.	log rollover test, corruption injection and event-timestamp review

Verification, margin and future-change rule

The release baseline remains valid across log schema, NVM, retention policy, timestamp source or service tool only within demonstrated similarity. Otherwise, log rollover test, corruption injection and event-timestamp review is repeated or updated.

USER ALARMS, PRIORITY & POWER-STATE INDICATION

User Alarms, Priority & Power-State Indication is controlled as a distinct product mechanism. Power alarms shall distinguish mains loss, battery use, low reserve, pack fault, transfer failure and pump-rail abnormality with program-defined priority.

The credible failure concern is alarm indication is stale, contradictory or unavailable. The allocated response is to default to the conservative high-priority power status and preserve diagnosable state.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-091	Power alarms shall distinguish mains loss, battery use, low reserve, pack fault, transfer failure and pump-rail abnormality with program-defined priority.	alarm-state test, indicator fault injection and human-factors evidence as applicable
ECMOPDU-092	If alarm indication is stale, contradictory or unavailable, the system shall default to the conservative high-priority power status and preserve diagnosable state.	alarm-state test, indicator fault injection and human-factors evidence as applicable
ECMOPDU-093	A change to display, LED, buzzer, alarm priority, UI firmware or localization shall reopen alarm-state test, indicator fault injection and human-factors evidence as applicable before production release.	alarm-state test, indicator fault injection and human-factors evidence as applicable

Verification, margin and future-change rule

Future variants affecting display, LED, buzzer, alarm priority, UI firmware or localization are screened against the mechanism on this page. The release owner may retain prior evidence only where alarm-state test, indicator fault injection and human-factors evidence as applicable confirms the similarity boundary.

SOURCE-STATUS INDICATORS & OPERATOR DECISION SUPPORT

The released architecture assigns explicit ownership to source-status indicators & operator decision support. Operators shall be able to identify which sources are present, qualified, active and degraded without interpreting raw electrical values.

A release review shall consider source indication disagrees with actual arbitration state; normal service is not assumed when that condition is present, and the product must raise a diagnostic/alarm discrepancy and use controller-internal qualified state for power decisions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-094	Operators shall be able to identify which sources are present, qualified, active and degraded without interpreting raw electrical values.	UI/controller correlation test and source-disagreement injection
ECMOPDU-095	Detection of source indication disagrees with actual arbitration state shall cause the system to raise a diagnostic/alarm discrepancy and use controller-internal qualified state for power decisions.	UI/controller correlation test and source-disagreement injection
ECMOPDU-096	Evidence reuse after status display, state mapping, source-good logic or UI communication is permitted only when UI/controller correlation test and source-disagreement injection demonstrates that the affected assumptions remain valid.	UI/controller correlation test and source-disagreement injection

Verification, margin and future-change rule

The change trigger for this page is status display, state mapping, source-good logic or UI communication. The impact record identifies moved assumptions and the exact UI/controller correlation test and source-disagreement injection needed for closure.

AMBULANCE / HELICOPTER POWER TRANSIENTS & TRANSPORT SUPPLY

For ambulance / helicopter power transients & transport supply, the design intent is translated into an auditable interface and state model. The PDU shall tolerate the approved ambulance/helicopter and facility power conditions defined by the transport program without losing pump continuity.

Verification focuses on the point where transport supply is interrupted or produces an out-of-envelope transient. The required containment is to reject the invalid source and continue from qualified battery power while capturing the event.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-097	The PDU shall tolerate the approved ambulance/helicopter and facility power conditions defined by the transport program without losing pump continuity.	transport power DVP, transient injection and source-transfer verification
ECMOPDU-098	The design shall detect transport supply is interrupted or produces an out-of-envelope transient and shall reject the invalid source and continue from qualified battery power while capturing the event.	transport power DVP, transient injection and source-transfer verification
ECMOPDU-099	Vehicle/aircraft adapter, mains supply, input filter, surge protection or source profile shall be treated as a controlled engineering change; approval requires transport power DVP, transient injection and source-transfer verification.	transport power DVP, transient injection and source-transfer verification

Verification, margin and future-change rule

Changes involving vehicle/aircraft adapter, mains supply, input filter, surge protection or source profile are engineering events, not editorial updates; approval depends on the targeted transport power DVP, transient injection and source-transfer verification listed here.

HIGH-IMPACT CHASSIS, SHOCK & DROP SURVIVAL

This page closes the product-level behavior for high-impact chassis, shock & drop survival. The chassis and internal battery/power assemblies shall remain mechanically secure and electrically safe after the approved transport shock/drop events.

The mechanism is incomplete unless it addresses impact causes connector, battery, PCB or isolation damage beyond accepted limits and can declare the unit unserviceable or enter the allocated fault state rather than continuing silently without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-100	The chassis and internal battery/power assemblies shall remain mechanically secure and electrically safe after the approved transport shock/drop events.	drop/shock qualification, post-event electrical inspection and functional retest
ECMOPDU-101	Loss or corruption involving impact causes connector, battery, PCB or isolation damage beyond accepted limits shall not permit silent normal operation; the system shall declare the unit unserviceable or enter the allocated fault state rather than continuing silently.	drop/shock qualification, post-event electrical inspection and functional retest
ECMOPDU-102	Production equivalence following chassis, battery restraint, connector, PCB mount, handle or fastener requires closure of drop/shock qualification, post-event electrical inspection and functional retest.	drop/shock qualification, post-event electrical inspection and functional retest

Verification, margin and future-change rule

For chassis, battery restraint, connector, PCB mount, handle or fastener, the configuration owner documents evidence reuse, new risks and the required drop/shock qualification, post-event electrical inspection and functional retest before sign-off.

IP54 ENCLOSURE, SEALING & FLUID EXPOSURE

Ip54 Enclosure, Sealing & Fluid Exposure is treated as an engineering boundary rather than a descriptive feature. The enclosure shall meet the baseline IP54 mission for the released seals, connectors and service-access configuration.

Seal damage or fluid ingress threatens electrical safety or power continuity is treated as a diagnosable abnormal condition. The system response is to detect where allocated, remove affected unit from service and prevent unqualified continued use.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-103	The enclosure shall meet the baseline IP54 mission for the released seals, connectors and service-access configuration.	ingress test, seal inspection and post-ingress electrical verification
ECMOPDU-104	When seal damage or fluid ingress threatens electrical safety or power continuity is present, the product shall detect where allocated, remove affected unit from service and prevent unqualified continued use.	ingress test, seal inspection and post-ingress electrical verification
ECMOPDU-105	The released configuration shall not be extended across seal, connector, gasket, vent, service door or enclosure revision without ingress test, seal inspection and post-ingress electrical verification.	ingress test, seal inspection and post-ingress electrical verification

Verification, margin and future-change rule

Any modification to seal, connector, gasket, vent, service door or enclosure revision is accepted only after the affected assumptions are rechecked and ingress test, seal inspection and post-ingress electrical verification is closed.

TRANSPORT THERMAL / ALTITUDE / ORIENTATION ENVELOPE

The product baseline allocates measurable behavior to transport thermal / altitude / orientation envelope. The power architecture shall be verified across the approved ICU and transport thermal, altitude and orientation conditions that affect batteries, cooling and source transfer.

The design prevents silent continuation when environmental condition exceeds the released operating envelope; it must instead maintain pump power if safely possible, shed nonessential loads and issue the appropriate environment/power alarm.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-106	The power architecture shall be verified across the approved ICU and transport thermal, altitude and orientation conditions that affect batteries, cooling and source transfer.	environmental DVP, altitude/thermal test and orientation/load review
ECMOPDU-107	The released implementation shall maintain pump power if safely possible, shed nonessential loads and issue the appropriate environment/power alarm whenever environmental condition exceeds the released operating envelope.	environmental DVP, altitude/thermal test and orientation/load review
ECMOPDU-108	When ambient range, altitude, orientation, cooling path, battery or transport enclosure occurs, the configuration record shall identify affected assumptions and close environmental DVP, altitude/thermal test and orientation/load review.	environmental DVP, altitude/thermal test and orientation/load review

Verification, margin and future-change rule

The release baseline remains valid across ambient range, altitude, orientation, cooling path, battery or transport enclosure only within demonstrated similarity. Otherwise, environmental DVP, altitude/thermal test and orientation/load review is repeated or updated.

CONNECTOR RETENTION, CABLE STRAIN & PUMP-POWER HARNESS

Connector Retention, Cable Strain & Pump-Power Harness is controlled as a distinct product mechanism. Pump-power and source connectors shall have mechanical retention and strain relief appropriate to transport motion and repeated clinical handling.

The credible failure concern is connector unlatch, high contact resistance or harness strain is detected or demonstrated. The allocated response is to prevent release/service of the affected assembly and preserve alternate-source capability where applicable.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-109	Pump-power and source connectors shall have mechanical retention and strain relief appropriate to transport motion and repeated clinical handling.	retention/strain test, contact-resistance measurement and post-stress DVP
ECMOPDU-110	If connector unlatch, high contact resistance or harness strain is detected or demonstrated, the system shall prevent release/service of the affected assembly and preserve alternate-source capability where applicable.	retention/strain test, contact-resistance measurement and post-stress DVP
ECMOPDU-111	A change to connector, latch, cable, strain relief, pin plating or harness routing shall reopen retention/strain test, contact-resistance measurement and post-stress DVP before production release.	retention/strain test, contact-resistance measurement and post-stress DVP

Verification, margin and future-change rule

Future variants affecting connector, latch, cable, strain relief, pin plating or harness routing are screened against the mechanism on this page. The release owner may retain prior evidence only where retention/strain test, contact-resistance measurement and post-stress DVP confirms the similarity boundary.

MANUFACTURING TRACEABILITY & CONFIGURATION GENEALOGY

The released architecture assigns explicit ownership to manufacturing traceability & configuration genealogy. Each PDU shall be traceable to hardware, firmware, batteries, mains module, critical power components and EOL test records.

A release review shall consider genealogy or critical component identity is incomplete; normal service is not assumed when that condition is present, and the product must quarantine the unit until the record is reconciled and release evidence is complete.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-112	Each PDU shall be traceable to hardware, firmware, batteries, mains module, critical power components and EOL test records.	MES audit, serialization test and configuration reconciliation
ECMOPDU-113	Detection of genealogy or critical component identity is incomplete shall cause the system to quarantine the unit until the record is reconciled and release evidence is complete.	MES audit, serialization test and configuration reconciliation
ECMOPDU-114	Evidence reuse after BOM, battery lot, firmware, mains module, manufacturing site or test database is permitted only when MES audit, serialization test and configuration reconciliation demonstrates that the affected assumptions remain valid.	MES audit, serialization test and configuration reconciliation

Verification, margin and future-change rule

The change trigger for this page is BOM, battery lot, firmware, mains module, manufacturing site or test database. The impact record identifies moved assumptions and the exact MES audit, serialization test and configuration reconciliation needed for closure.

BATTERY PACK IDENTITY, PAIRING & REPLACEMENT TRACEABILITY

For battery pack identity, pairing & replacement traceability, the design intent is translated into an auditable interface and state model. Installed packs shall be uniquely identified and associated with cycle/impedance/service history so replacement does not inherit incorrect aging data.

Verification focuses on the point where pack identity is duplicated, unreadable or mismatched with stored history. The required containment is to treat health history as invalid and require authorized service disposition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-115	Installed packs shall be uniquely identified and associated with cycle/impedance/service history so replacement does not inherit incorrect aging data.	pack-swap test, identity mismatch cases and service record audit
ECMOPDU-116	The design shall detect pack identity is duplicated, unreadable or mismatched with stored history and shall treat health history as invalid and require authorized service disposition.	pack-swap test, identity mismatch cases and service record audit
ECMOPDU-117	Battery ID, pack electronics, NVM, service tool or replacement workflow shall be treated as a controlled engineering change; approval requires pack-swap test, identity mismatch cases and service record audit.	pack-swap test, identity mismatch cases and service record audit

Verification, margin and future-change rule

Changes involving battery ID, pack electronics, NVM, service tool or replacement workflow are engineering events, not editorial updates; approval depends on the targeted pack-swap test, identity mismatch cases and service record audit listed here.

EOL HOT-SWAP & UNINTERRUPTED SWITCHOVER VERIFICATION

This page closes the product-level behavior for eol hot-swap & uninterrupted switchover verification. EOL shall prove source removal/insertion and mains/battery transfer with the representative pump load while monitoring interruption and rail deviation.

The mechanism is incomplete unless it addresses switchover exceeds the released pump tolerance or produces unstable source chatter and can reject the unit or affected design variant until the transfer mechanism is corrected without relying on an unrelated convenience function.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-118	EOL shall prove source removal/insertion and mains/battery transfer with the representative pump load while monitoring interruption and rail deviation.	EOL switchover waveform, pump-load test and anti-chatter evidence
ECMOPDU-119	Loss or corruption involving switchover exceeds the released pump tolerance or produces unstable source chatter shall not permit silent normal operation; the system shall reject the unit or affected design variant until the transfer mechanism is corrected.	EOL switchover waveform, pump-load test and anti-chatter evidence
ECMOPDU-120	Production equivalence following test load, source emulator, OR-ing path, thresholds or firmware requires closure of EOL switchover waveform, pump-load test and anti-chatter evidence.	EOL switchover waveform, pump-load test and anti-chatter evidence

Verification, margin and future-change rule

For test load, source emulator, OR-ing path, thresholds or firmware, the configuration owner documents evidence reuse, new risks and the required EOL switchover waveform, pump-load test and anti-chatter evidence before sign-off.

EOL LOAD, THERMAL, ISOLATION & ALARM TEST

Eol Load, Thermal, Isolation & Alarm Test is treated as an engineering boundary rather than a descriptive feature. EOL/qualification shall verify pump rail, secondary-load isolation, critical alarms and applicable electrical-safety tests for the exact unit configuration.

Required load, thermal, alarm or safety evidence is missing or outside limits is treated as a diagnosable abnormal condition. The system response is to block unit/variant release until objective evidence is closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-121	EOL/qualification shall verify pump rail, secondary-load isolation, critical alarms and applicable electrical-safety tests for the exact unit configuration.	EOL functional records, thermal spot checks, dielectric/leakage tests and alarm verification
ECMOPDU-122	When required load, thermal, alarm or safety evidence is missing or outside limits is present, the product shall block unit/variant release until objective evidence is closed.	EOL functional records, thermal spot checks, dielectric/leakage tests and alarm verification
ECMOPDU-123	The released configuration shall not be extended across test fixture, acceptance limit, isolation path, sensor or manufacturing process without EOL functional records, thermal spot checks, dielectric/leakage tests and alarm verification.	EOL functional records, thermal spot checks, dielectric/leakage tests and alarm verification

Verification, margin and future-change rule

Any modification to test fixture, acceptance limit, isolation path, sensor or manufacturing process is accepted only after the affected assumptions are rechecked and EOL functional records, thermal spot checks, dielectric/leakage tests and alarm verification is closed.

SINGLE-FAULT / SOURCE-LOSS DVP

The product baseline allocates measurable behavior to single-fault / source-loss dvp. DVP shall inject representative source, OR-ing, charger, sensor, controller and distribution faults while monitoring uninterrupted pump-power objectives.

The design prevents silent continuation when a single injected fault removes all pump power or defeats required alarm/containment; it must instead fail the safety release gate and revise the architecture before production approval.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-124	DVP shall inject representative source, OR-ing, charger, sensor, controller and distribution faults while monitoring uninterrupted pump-power objectives.	single-fault matrix, pump continuity waveforms and risk-control traceability
ECMOPDU-125	The released implementation shall fail the safety release gate and revise the architecture before production approval whenever a single injected fault removes all pump power or defeats required alarm/containment.	single-fault matrix, pump continuity waveforms and risk-control traceability
ECMOPDU-126	When fault model, source path, shared controller, sensor, connector or protection occurs, the configuration record shall identify affected assumptions and close single-fault matrix, pump continuity waveforms and risk-control traceability.	single-fault matrix, pump continuity waveforms and risk-control traceability

Verification, margin and future-change rule

The release baseline remains valid across fault model, source path, shared controller, sensor, connector or protection only within demonstrated similarity. Otherwise, single-fault matrix, pump continuity waveforms and risk-control traceability is repeated or updated.

TRANSPORT, CONTINUOUS-OPERATION & BATTERY-ENDURANCE DVP

Transport, Continuous-Operation & Battery-Endurance Dvp is controlled as a distinct product mechanism. System verification shall combine transport power loss, battery aging, representative pump load and environmental conditions over the approved mission duration.

The credible failure concern is mission test reveals insufficient battery endurance, thermal margin or source-transfer robustness. The allocated response is to keep the release gate open and update the controlled parameter/architecture baseline.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-127	System verification shall combine transport power loss, battery aging, representative pump load and environmental conditions over the approved mission duration.	mission-profile DVP, aged-battery tests and continuous-load thermal data
ECMOPDU-128	If mission test reveals insufficient battery endurance, thermal margin or source-transfer robustness, the system shall keep the release gate open and update the controlled parameter/architecture baseline.	mission-profile DVP, aged-battery tests and continuous-load thermal data
ECMOPDU-129	A change to pump load, battery capacity, transport profile, environment or service interval shall reopen mission-profile DVP, aged-battery tests and continuous-load thermal data before production release.	mission-profile DVP, aged-battery tests and continuous-load thermal data

Verification, margin and future-change rule

Future variants affecting pump load, battery capacity, transport profile, environment or service interval are screened against the mechanism on this page. The release owner may retain prior evidence only where mission-profile DVP, aged-battery tests and continuous-load thermal data confirms the similarity boundary.

15-YEAR LIFECYCLE, PACK REPLACEMENT & OBSOLESCENCE

The released architecture assigns explicit ownership to 15-year lifecycle, pack replacement & obsolescence. The 15-year lifecycle plan shall track battery cycles/impedance, mandatory pack replacement, critical component aging, service parts and firmware/tool support.

A release review shall consider battery, component or support dependency reaches its approved service limit; normal service is not assumed when that condition is present, and the product must trigger planned replacement/obsolescence action before pump-power reliability is compromised.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-130	The 15-year lifecycle plan shall track battery cycles/impedance, mandatory pack replacement, critical component aging, service parts and firmware/tool support.	lifecycle plan, replacement criteria, obsolescence review and field-data governance
ECMOPDU-131	Detection of battery, component or support dependency reaches its approved service limit shall cause the system to trigger planned replacement/obsolescence action before pump-power reliability is compromised.	lifecycle plan, replacement criteria, obsolescence review and field-data governance
ECMOPDU-132	Evidence reuse after battery supplier, OR-ing device, charger, MCU, NVM, service tool or firmware support is permitted only when lifecycle plan, replacement criteria, obsolescence review and field-data governance demonstrates that the affected assumptions remain valid.	lifecycle plan, replacement criteria, obsolescence review and field-data governance

Verification, margin and future-change rule

The change trigger for this page is battery supplier, OR-ing device, charger, MCU, NVM, service tool or firmware support. The impact record identifies moved assumptions and the exact lifecycle plan, replacement criteria, obsolescence review and field-data governance needed for closure.

FINAL RELEASE, CLASSIFICATION CLARIFICATION & SUPPLIER EQUIVALENCE

For final release, classification clarification & supplier equivalence, the design intent is translated into an auditable interface and state model. Final release shall resolve the baseline phrase IEC 60601-1 Class III compliant into the correct program regulatory/electrical classification and close all pump-continuity, electrical-safety, manufacturing and lifecycle evidence.

Verification focuses on the point where classification, supplier or variant evidence is incomplete or contradictory. The required containment is to withhold production-equivalence approval until the regulatory mapping and affected analysis/regression are closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
ECMOPDU-133	Final release shall resolve the baseline phrase IEC 60601-1 Class III compliant into the correct program regulatory/electrical classification and close all pump-continuity, electrical-safety, manufacturing and lifecycle evidence.	release checklist, regulatory classification record, supplier change review and evidence index
ECMOPDU-134	The design shall detect classification, supplier or variant evidence is incomplete or contradictory and shall withhold production-equivalence approval until the regulatory mapping and affected analysis/regression are closed.	release checklist, regulatory classification record, supplier change review and evidence index
ECMOPDU-135	Regulatory classification, medical safety standard profile, supplier, BOM, firmware or manufacturing site shall be treated as a controlled engineering change; approval requires release checklist, regulatory classification record, supplier change review and evidence index.	release checklist, regulatory classification record, supplier change review and evidence index

Verification, margin and future-change rule

Changes involving regulatory classification, medical safety standard profile, supplier, BOM, firmware or manufacturing site are engineering events, not editorial updates; approval depends on the targeted release checklist, regulatory classification record, supplier change review and evidence index listed here.